Home Theatre

Sample Client

12 Example Street, Sampleton, CA 90000, US

DRAWING AV-100 · PROJECT RM-2026-001 · REVISION A · ISSUED 2026-07-18

INTEGRATOR

Example Integration Co.

SIGN-OFF

DRAWN RackPal · CHECKED — · APPROVED —

DRAWING INDEX

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RACKS

RACK	SIZE	DEVICES	RU USED	WEIGHT	POWER	HEAT
RK-1 Cinema Rack	24U	12	F 18 · R 13	176.6 lb	≥ 1673 VA (1 unknown)	≥ 4436 BTU/hr (1 unknown)

REV	DATE	DESCRIPTION	BY
A	2026-03-09	Issued for review	RackPal

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COVER AND DRAWING INDEX

DRAWN	RackPal	SCALE	NTS	SIZE	B	DATE	2026-07-18	REV	A
CHK'D	—	PROJ	RM-2026-001	DISC	T	FILE	home_theatre.rppj	SHEET	1/10

Example Integration Co.

SYMBOL LEGEND: SEE SHEET T0.01

CABLE TYPES

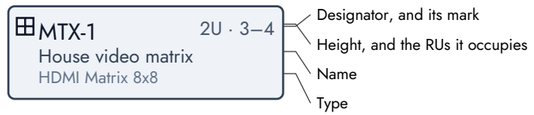
Colour marks the signal domain, not the cable type: no palette stays distinguishable to colour-blind readers past about eight hues.
Dash, weight, mark and tag carry the type, so every type below still reads in greyscale (AC-054).

TYPE	FAMILY	IN COLOUR	IN GREYSCALE
Analog RCA	Line level, unbalanced (RCA)	● AUD·AUD	● AUD·AUD
HDMI	HDMI 2.0 (18 Gbps)	▲ VID·HD20	▲ VID·HD20
	HDMI 2.1 (48 Gbps)	▲ VID·HD21	▲ VID·HD21
	HDMI 1.4	▲ VID·HD14	▲ VID·HD14
Network	Ethernet (copper)	■ NET·NET	■ NET·NET
	PoE / PoE+ / PoE++	■ NET·PoE	■ NET·PoE
IR	IR emitter (flasher)	✚ CTL·IR	✚ CTL·IR
	IR receiver (powered eye)	✚ CTL·IRRX	✚ CTL·IRRX
	IR direct (device input)	✚ CTL·IRD	✚ CTL·IRD
RS-232	RS-232	✚ CTL·232	✚ CTL·232
RG6	RF / CATV distribution	▽ RF·RF	▽ RF·RF
	Satellite / SWM	▽ RF·SAT	▽ RF·SAT
	OTA antenna	▽ RF·OTA	▽ RF·OTA
S/PDIF	S/PDIF, coaxial (75 Ω)	◆ DIG·SPD	◆ DIG·SPD
TOSLINK	TOSLINK (optical S/PDIF)	◇ DIG·OPT	◇ DIG·OPT
Speaker level	Speaker level, low-Z	◖ AUD·SPK	◖ AUD·SPK
	Speaker, constant-voltage 70 / 100 / 25 V	○ AUD·70V	○ AUD·70V
Power	AC branch circuit	⬢ PWR·PWR	⬢ PWR·PWR
	IEC device power (C13/C14, C19/C20)	⬢ PWR·IEC	⬢ PWR·IEC

SYMBOLS

- Matrix switch
- Switcher
- Processor
- Patch panel
- Blank or vent
- Distribution amplifier
- Amplifier
- Source
- Power
- Shelf

DEVICE BLOCK



An example block, not a device in this project.

CONVENTIONS

- Border: NEW
- Hatched: a blank, vent or shelf
- 2U · 3–4 — Height, and the RUs a device occupies
- 176.6 lb — Load gutter: weight carried from this RU up

IDENTIFIERS

Device — {TYPE_PREFIX}{SEP}{SEQ} → AMP-1
- TYPE_PREFIX = AMP, prefix of the device type
- SEP = -, separator
- SEQ = 1, sequence number

Port — J{PORT_SEQ:2} → J03
- PORT_SEQ = 3, port sequence on the device

Cable — W{SEQ:3} → W001
- SEQ = 1, sequence number

Outlet, field end — {ROOM_ABBR}-{SERVICE}-{SEQ} → LR-TV-1
- ROOM_ABBR = LR, room abbreviation
- SERVICE = TV, service at the outlet
- SEQ = 1, sequence number

Outlet, rack end — {FLOOR}{SPACE}-{PANEL}{PORT:2} → 1A-A12
- FLOOR = 1, floor
- SPACE = A, space on the floor
- PANEL = A, patch panel letter
- PORT = 12, panel port number

ABBREVIATIONS AND NOTES

RU	rack unit, 1.75 in
NTS	not to scale
REV	revision
CHK'D	checked by
DISC	discipline
PROJ	project number
AUD	Analog Audio (a tag prefix)
CTL	Control (a tag prefix)
DIG	Digital Audio (a tag prefix)
NET	Network (a tag prefix)
PWR	Power (a tag prefix)
RF	Analog Video & RF (a tag prefix)
STR	Structure (a tag prefix)
VID	Digital Video (a tag prefix)

Do not scale a sheet marked NTS.

A value shown as — is not known, and was not guessed.

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SYMBOL LEGEND

DRAWN	RackPal	SCALE	NTS	SIZE	B	DATE	2026-07-18	REV	A
CHK'D	—	PROJ	RM-2026-001	DISC	T	FILE	home_theatre.rppj	SHEET	2/10

Example Integration Co.

PROJECT-WIDE EXCEPTIONS (0)

RULE	CHANGE	REASON	BY	DATE

None. Every rule runs at its catalogue severity, and none is muted.

OPEN FINDINGS (4)

SEVERITY	RULE	LOCUS	MESSAGE
WARNING (4)			
WARNING	RCK-013	SW-1	SW-1 is 7.7 kg on front ears rated to 7.0 kg. It needs rear rail support or a shelf in RK-1, and may require adjustment of the rack ears.
WARNING	RCK-013	AMP-1	AMP-1 is 15.4 kg on front ears rated to 7.0 kg. It needs rear rail support or a shelf in RK-1, and may require adjustment of the rack ears.
WARNING	RCK-013	MTX-1	MTX-1 is 14.1 kg on front ears rated to 7.0 kg. It needs rear rail support or a shelf in RK-1, and may require adjustment of the rack ears.
WARNING	RCK-013	SEQ-1	SEQ-1 is 8.2 kg on front ears rated to 7.0 kg. It needs rear rail support or a shelf in RK-1, and may require adjustment of the rack ears.

ACKNOWLEDGED FINDINGS (0)

SEVERITY	RULE	LOCUS	MESSAGE	REASON	BY	DATE

None.

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EXCEPTIONS AND ACKNOWLEDGEMENTS

DRAWN	RackPal	SCALE	NTS	SIZE	B	DATE	2026-07-18	REV	A
CHK'D	—	PROJ	RM-2026-001	DISC	T	FILE	home_theatre.rppj	SHEET	3/10

Example Integration Co.

SYMBOL LEGEND: SEE SHEET T0.01

SCALE APPLIES TO THE MOUNTING PANEL. RULER AND GUTTER ARE ANNOTATION.
RACK GROUND BAR PRESENT. BOND TO BUILDING GROUND PER LOCAL CODE.

RACK TOTAL 176.6 lb
CAPACITY 1102.3 lb
DRAW ≥ 1673 VA (1 unknown)
HEAT ≥ 4436 BTU/hr (1 unknown)
NOT DRAWN HERE 1 OF 15 — SEE T6.30; 3 COUNTED AT TOP

1 ft

REV	DATE	DESCRIPTION	BY
A	2026-03-09	Issued for review	RackPal

Home Theatre — RK-1 Cinema Rack
12 Example Street, Sampleton, CA 90000, US
CLIENT Sample Client

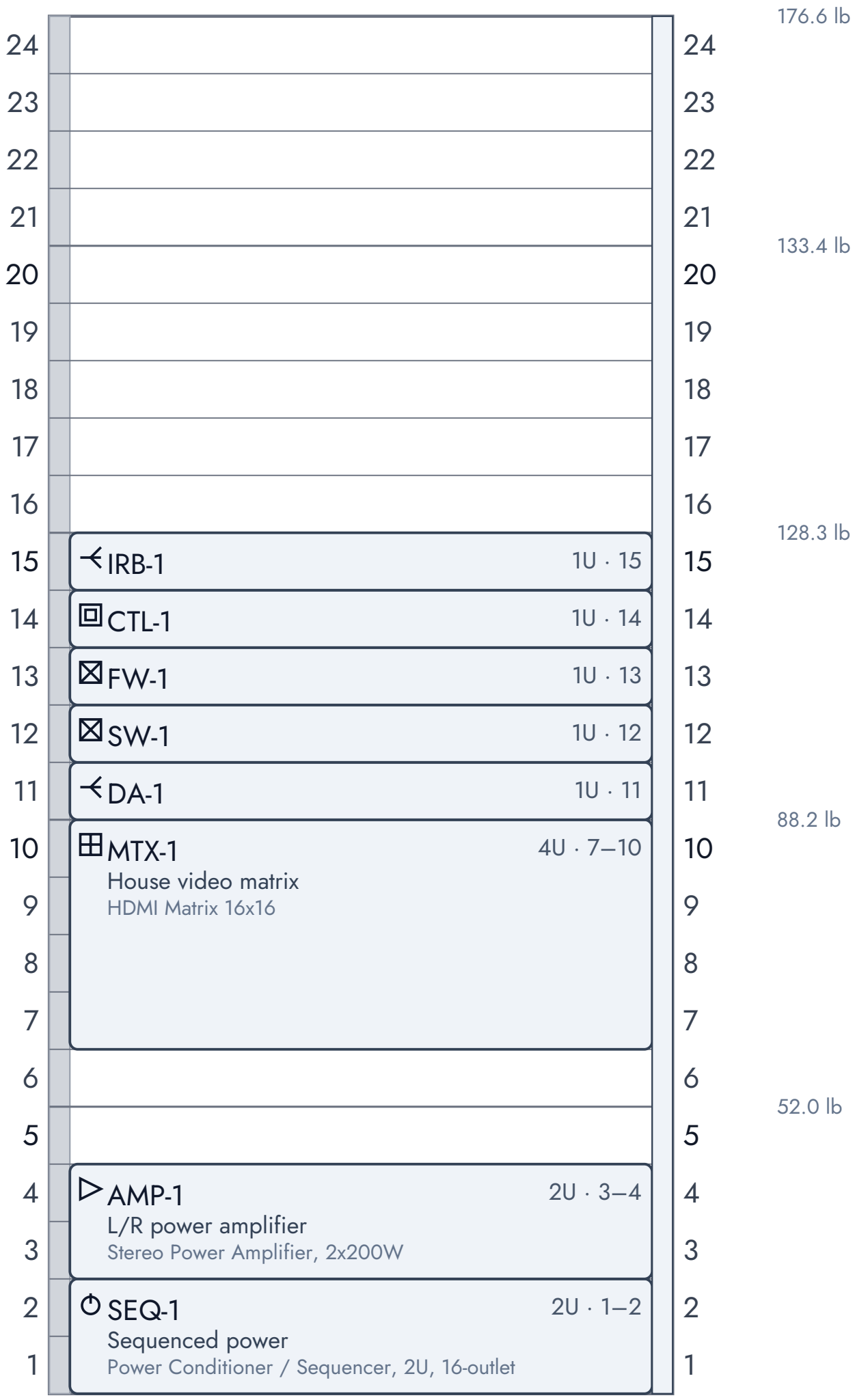
RACK ELEVATION — FRONT
RK-1 · Cinema Rack · 24U

DRAWN	RackPal	SCALE	1:4	SIZE	B	DATE	2026-07-18	REV	A
CHK'D	—	PROJ	RM-2026-001	DISC	T	FILE	home_theatre.rppj	SHEET	4/10

Example Integration Co.
SYMBOL LEGEND: SEE SHEET T0.01
SHEET 4 OF 10 · RACK ELEVATION — FRONT
SHEET NO. T2.01

REAR FACES SHOWN. PORT AND CONNECTION DETAIL: SEE T6.15.
REAR VIEW — MIRRORED
SCALE APPLIES TO THE MOUNTING PANEL. RULER AND GUTTER ARE ANNOTATION.
RACK GROUND BAR PRESENT. BOND TO BUILDING GROUND PER LOCAL CODE.

RACK TOTAL 176.6 lb
CAPACITY 1102.3 lb
DRAW ≥ 1673 VA (1 unknown)
HEAT ≥ 4436 BTU/hr (1 unknown)
NOT DRAWN HERE 7 OF 15 — 6 ON FRONT, 1 ONLY ON T6.30; 3 COUNTED AT TOP



1 ft

REV	DATE	DESCRIPTION	BY
A	2026-03-09	Issued for review	RackPal

Home Theatre — RK-1 Cinema Rack
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RACK ELEVATION — REAR
RK-1 · Cinema Rack · 24U

DRAWN	RackPal	SCALE	1:4	SIZE	B	DATE	2026-07-18	REV	A
CHK'D	—	PROJ	RM-2026-001	DISC	T	FILE	home_theatre.rppj	SHEET	5/10

Example Integration Co.
SYMBOL LEGEND: SEE SHEET T0.01
SHEET 5 OF 10 · RACK ELEVATION — REAR
SHEET NO. T2.02

PULL SHEET (0 runs)

CABLE ID	TYPE	FIELD LABEL	RACK LABEL	FROM — ROOM / OUTLET	TO — RACK / RU / FACE	MEDIA	LEN	PATHW…	SIGN-OFF

This project has no cables yet.

TERMINATION SCHEDULE (0 runs)

CABLE ID	TYPE	RACK LABEL	FIELD LABEL	RACK / RU / FACE	DEVICE	PORT	FAR END	FAR PORT	MEDIA	FIELD TERM	RACK TERM	CONDUCT... / NOTES

This project has no cables yet.

EQUIPMENT AND RACK MATERIALS. CABLE SCHEDULES: SEE T6.10 AND T6.15. POWER SCHEDULE NOT ISSUED IN THIS SET. 14 DEVICES NOT IN A RACK.

QTY	DESIGNATOR	MANUFACTURER	MODEL	DESCRIPTION	RU	RACK	LOCATION	FACE	MASS (LB)	DRAW (VA)	LIFECYCLE	NOTES
RK-1 · Cinema Rack · 24U												
1	PP-1	Generic	PP-24	Room patch panel	1U · 17	RK-1	Equipment Closet	FRONT	3.1	—	NEW	
1	IRB-1	Generic	IRB-4X16	IR distribution	1U · 15	RK-1	Equipment Closet	FRONT	2.0	7	NEW	
1	CTL-1	Generic	CTL-PROC-1	Control processor	1U · 14	RK-1	Equipment Closet	FRONT	7.1	32	NEW	
1	FW-1	Generic	FW-RTR-1	Router / firewall	1U · 13	RK-1	Equipment Closet	FRONT	9.0	58	NEW	
1	SW-1	Generic	SW-48-POE	AV network switch	1U · 12	RK-1	Equipment Closet	FRONT	17.0	400	NEW	
1	DA-1	Generic	DA-1X8-HDMI	Projector feed DA	1U · 11	RK-1	Equipment Closet	FRONT	5.1	17	NEW	
1	MTX-1	Generic	MTX-16X16-HDMI	House video matrix	4U · 7–10	RK-1	Equipment Closet	FRONT	31.1	116	NEW	
1	AMP-1	Generic	AMP-2X200	L/R power amplifier	2U · 3–4	RK-1	Equipment Closet	FRONT	34.0	585	NEW	
1	SEQ-1	Generic	SEQ-2U-16	Sequenced power	2U · 1–2	RK-1	Equipment Closet	FRONT	18.1	12	NEW	
1	AVR-1	Generic	AVR-7.2CH	Cinema AV receiver	—	RK-1	Equipment Closet	SHELF	26.0	412	NEW	
1	STB-1	Generic	STB-1	Cable box	—	RK-1	Equipment Closet	SHELF	3.1	24	NEW	
1	PDU-1	Generic	PDU-0U-24	Rear PDU	0U	RK-1	Equipment Closet	0U	14.1	10	NEW	
							RK-1 SUBTOTAL	12 DEVICES + 3 PASSIVES (SEE MATERIALS)			176.6 lb	≥ 1673 VA (1 unknown)
NOT IN A RACK												
1	PROJ-1	Generic	PROJ-1	Cinema projector	—	—	Media Room	NONE	16.1	295	NEW	Ceiling mount, rear of room
1	SCRN-1	Generic	SCRN-MOTOR-1	Motorised screen	—	—	Media Room	NONE	50.0	100	NEW	Recessed, front wall
1	DISP-1	Generic	DISP-1	Secondary display	—	—	Media Room	NONE	39.9	126	NEW	Side wall, casual viewing
1	SPK-1	Generic	SPK-WALL-1	Front left	—	—	Media Room	NONE	7.1	—	NEW	In-wall, behind screen
1	SPK-2	Generic	SPK-WALL-1	Centre	—	—	Media Room	NONE	7.1	—	NEW	In-wall, behind screen
1	SPK-3	Generic	SPK-WALL-1	Front right	—	—	Media Room	NONE	7.1	—	NEW	In-wall, behind screen
1	SPK-4	Generic	SPK-CEIL-PAIR	Surround pair	—	—	Media Room	NONE	6.0	—	NEW	In-ceiling, side walls
1	SPK-5	Generic	SPK-CEIL-PAIR	Height pair	—	—	Media Room	NONE	6.0	—	NEW	In-ceiling, front
1	SUB-1	Generic	SUB-POWERED-1	Subwoofer	—	—	Media Room	NONE	50.0	231	NEW	Front left corner
1	TP-1	Generic	TP-1	Room touchpanel	—	—	Media Room	NONE	3.1	—	NEW	In-wall, entry door
1	IRR-1	Generic	IRR-EYE-1	IR receiver	—	—	Media Room	NONE	0.1	—	NEW	Under screen trim
1	KP-1	Generic	KP-1	Hall keypad	—	—	Hallway	NONE	0.4	—	NEW	Beside media room door
1	AP-1	Generic	AP-1	Hall access point	—	—	Hallway	NONE	2.0	—	NEW	Ceiling, centre of hall
1	ONT-1	Generic	ONT-1	Fibre ONT	—	—	Equipment Closet	NONE	2.4	—	NEW	Wall beside rack
								NOT IN A RACK SUBTOTAL		14 DEVICES	197.2 lb	≥ 752 VA (10 unknown)

EQUIPMENT AND RACK MATERIALS. CABLE SCHEDULES: SEE T6.10 AND T6.15. POWER SCHEDULE NOT ISSUED IN THIS SET. 14 DEVICES NOT IN A RACK.

QTY	DESIGNATOR	MANUFACTURER	MODEL	DESCRIPTION	RU	RACK	LOCATION	FACE	MASS (LB)	DRAW (VA)	LIFECYCLE	NOTES
MATERIALS												
1		Generic	BLK-1U	Blank Panel, 1U	RK-1 × 1				1.0	0	NEW	
1		Generic	SHF-2U	Rack Shelf, 2U, vented	RK-1 × 1				5.1	0	NEW	
1		Generic	VNT-1U	Vent Panel, 1U, 64% open	RK-1 × 1				1.0	0	NEW	

CABLE FOOTAGE BY MEDIA

MEDIA	RUNS	ESTIMATED	LENGTH UNKNOWN	FOOTAGE	WITH WASTE (+10%)
NO FIELD CABLE IN THIS PROJECT.					

HARDWARE IMPLIED BY THE RULES

QTY	ITEM	FOR	RACK	RULE	NOTES
1	Rear rail support	AMP-1	RK-1	RCK-013	
1	Rear rail support	MTX-1	RK-1	RCK-013	
1	Rear rail support	SEQ-1	RK-1	RCK-013	
1	Rear rail support	SW-1	RK-1	RCK-013	

JUMPERS AND POWER CORDS

QTY	MEDIA	LENGTH	COLOUR	NOTES
NO JUMPERS OR POWER CORDS IN THIS PROJECT.				

IN-LINE ADAPTERS

QTY	ADAPTER	MODEL
NO IN-LINE ADAPTER IS DECLARED.		

OUTLET MAP

PROVIDER	OUTLET	DEVICE	INLET	CORD	RACK
NO CORD IS DRAWN TO AN OUTLET.					

PDU-1: 0 of 24 outlets assigned, 24 free.

SEQ-1: 0 of 16 outlets assigned, 16 free.

18 devices have no cord drawn and are counted as one outlet per inlet.

POWERED IR RECEIVER PORTS (6)

A powered receiver port carries +12 V on the ring. Never plug a mono emitter into its jack: the plug bridges ring to sleeve.

DEVICE	PORT	VARIANT	PIN MAP
IRB-1	IR IN (from processor)	xantech-stereo	TIP IR · RING +12 V · SLEEVE GND
IRB-1	IR RX 1	xantech-stereo	TIP IR · RING +12 V · SLEEVE GND
IRB-1	IR RX 2	xantech-stereo	TIP IR · RING +12 V · SLEEVE GND
IRB-1	IR RX 3	xantech-stereo	TIP IR · RING +12 V · SLEEVE GND
IRB-1	IR RX 4	xantech-stereo	TIP IR · RING +12 V · SLEEVE GND
IRR-1	PIGTAIL TO BLOCK	xantech-stereo	TIP IR · RING +12 V · SLEEVE GND